

Summary Report

1. Introduction

This report presents a comprehensive analysis of hotel booking data with the objective of understanding customer booking behaviour, identifying cancellation trends, evaluating channel and property performance, and uncovering factors affecting revenue and profitability. The analysis incorporates booking patterns across room types, star ratings, booking channels, stay durations, lead times, refunds, and promotional activities.

The findings aim to support data-driven decision-making and provide actionable recommendations to improve customer retention, reduce cancellations, optimize pricing strategies, and enhance overall business performance.

2. Key Observations and Trends

2.1 Booking Patterns Across Channels

- Different booking channels contribute differently to total bookings and revenue generation.
- Online booking channels account for a significant share of bookings due to convenience and accessibility.
- Certain channels generate higher average booking values, indicating the presence of premium customers or higher-value bookings.

2.2 Room Type Analysis

- Standard rooms account for the highest booking volume due to affordability.
- Premium room categories generate higher average booking values and profitability.
- Cancellation rates vary across room types, suggesting differences in customer commitment and price sensitivity.

2.3 Star Rating Analysis

- Higher-rated properties generally achieve higher booking values and revenue.
- Premium properties attract customers willing to spend more per booking.
- Cancellation behavior differs across star categories, potentially due to varying customer expectations and travel plans.

2.4 Stay Length and Lead Time Trends

- Most bookings are concentrated within shorter stay durations.
- Customers with longer booking lead times tend to exhibit higher cancellation tendencies.
- Last-minute bookings generally show stronger booking commitment compared to bookings made well in advance.

2.5 Revenue and Profitability Trends

- Revenue generation varies across channels, room types, and property categories.
- Premium room categories contribute disproportionately to profitability despite lower booking volumes.
- Promotional discounts, cashback offers, and coupon redemptions influence booking values and margins.

3. Cancellation Analysis

3.1 Key Findings

- A notable proportion of bookings were cancelled, impacting revenue realization and occupancy planning.
- Cancellation rates differ across booking channels and room categories.
- Bookings with longer lead times show a greater tendency to be cancelled.
- Refund amounts are significantly associated with cancelled bookings.

3.2 Possible Root Causes

Several factors may contribute to cancellation behaviour:

1. Customers booking far in advance may alter travel plans before check-in.
2. Flexible cancellation policies can encourage speculative bookings.
3. Customers may find better deals through competing platforms.
4. Price-sensitive customers using discounts and promotional offers may have lower booking commitment.
5. Seasonal demand fluctuations and unforeseen travel disruptions can increase cancellations.

4. Root Cause Analysis

Why Some Channels Perform Better

- Online platforms provide easier access, broader reach, and better customer experience.
- Certain channels attract higher-value customers, leading to increased booking values.
- Marketing investments and channel-specific promotions influence booking volume and conversion rates.

Why Some Property Types Perform Better

- Premium hotels offer enhanced amenities and services, supporting higher pricing.
- Customer preference for quality experiences increases demand for higher-rated properties.
- Strong brand perception contributes to higher occupancy and customer loyalty.

Seasonal and Temporal Trends

- Booking activity varies throughout the year due to holidays, vacations, festivals, and business travel cycles.
- Peak travel periods contribute to increased booking values and occupancy levels.
- Off-season periods may experience lower demand and increased promotional activity.

5. Business Recommendations

5.1 Reduce Cancellations

- Introduce tiered cancellation policies with incentives for non-refundable bookings.
- Send automated reminders before check-in dates.
- Offer loyalty benefits for customers who complete bookings.
- Implement predictive models to identify high-risk cancellations and trigger retention campaigns.

5.2 Improve Profitability

- Focus marketing efforts on high-performing booking channels.
- Increase promotion of premium room categories with higher profit margins.
- Optimize cashback and coupon strategies to maintain profitability while driving conversions.
- Monitor channel-wise profit contribution and adjust commission structures accordingly.

5.3 Increase Repeat Bookings

- Launch loyalty and membership programs.
- Provide personalized offers based on customer booking history.
- Improve customer experience through targeted engagement and post-stay communication.
- Offer exclusive discounts for repeat customers.

5.4 Optimize Pricing and Promotions

- Implement dynamic pricing based on demand, seasonality, and occupancy levels.
- Use data-driven promotional campaigns during low-demand periods.
- Adjust room pricing based on customer segments and booking behavior.
- Evaluate the effectiveness of coupon and cashback programs to maximize return on investment.

6. Conclusion

The analysis highlights meaningful relationships between booking channels, room types, property ratings, customer behaviour, and profitability. Cancellation rates are influenced by lead time, booking channel, and pricing factors, while premium properties and high-value room categories contribute significantly to revenue generation.

By adopting targeted cancellation reduction strategies, optimizing pricing and promotional activities, and strengthening customer retention initiatives, the business can improve occupancy, profitability, and long-term customer value. The insights derived from this analysis provide a strong foundation for future operational and strategic decision-making.